Models, Math, and Real-World Signals

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Key Concepts

1) Signal-processing algorithms are usually derived on the basis of explicit or implicit mathematical models of signals.

2) Signal models represent prior knowledge about the characteristics of the signal, interference, or noise.

3) Models for physics-based applications often follow from the mathematical laws of physics.

4) Mathematical disciplines used in signal processing include calculus, probability, statistics, and linear algebra.

5) Deviation between models assumed for signals and the real signals can result in degraded performance.

6) Simpler models generally lead to less complex and more robust algorithms than highly detailed models.

7) Very detailed models can lead to outstanding performance when the model assumptions are correct.

Transcript

Models, Math, and Real-World Signals

Introduction

Perhaps the key tool of signal processing is mathematical modeling. The use of mathematical models for describing signal attributes is the basis for developing systematic approaches to design of signal-processing methods.

For example, we may model multiple measurements as a signal that repeats in noise that has a random character. This model suggests averaging the measurements to enhance the signal relative to the noise.

In a radar system we may wish to estimate the velocity of a plane relative to the radar. The "model" is based on the Doppler effect – objects moving toward the radar cause an increase in the frequency of the reflected signal, while those moving away cause a decrease. The magnitude of the frequency shift is proportional to object velocity. In this case the signal-processing algorithm estimates the frequency shift in the signal reflected from the plane, which is then converted to velocity.

Mathematical models for signals can be as diverse as signal processing applications. Indeed, identifying a model may require collaboration between an application expert and a signal-processing engineer. Once a model is defined, standard signal processing methodologies are usually applied to solve the problem.

Where do models come from?

Models for signals are often based on prior knowledge about the problem of interest, such as a partial characterization of the signal or noise. An example is the assumption the signal repeats in multiple measurements, while the noise is random.

Suppose we wish to reduce the blur in a picture due to camera motion. In this case we use a model for the effect of camera motion on the recorded image.

Physics

Models may also reflect the physics of the measurement system. For example, supposed multiple antennas are used to measure the signal reflected from a distant object in free space. The physics of plane-wave propagation tells us that the signals measured at different antennas are time-delayed versions of each other. The relative time delays are a function of the object location and antenna array geometry. The location dependent nature of the time delays can be used to separate the signals from objects at different locations.

Models based on physics are often relatively straightforward because the physical world is very well modeled by equations. For example, Maxwell's equations describe the interaction of electric and magnetic fields with physical

materials. They form the basis for signal models in radar, wireless communications, and remote sensing applications. The behavior of resonant cavities forms a basis for models of voiced (e.g., vowel sounds) speech. Statics tells us how force is distributed in a plate as a person walks. Laws of optics describe the distortion of an image caused by a telescope with an improperly ground mirror.

Biology

In contrast to physics-based applications, biology, as currently understood, does not lend itself as well to description by equations. Accurate signal-processing models for biological problems can be difficult to identify.

For example, the electrical signal generated by a local population of neurons in the cortex is often approximated as a current dipole in electroencephalography (EEG – measurement of electric fields at the scalp). The physics of the signal generated by a current dipole are well understood. However, in practice brain activity is due to many populations of neurons interacting in a complex and largely unknown manner. There are no generally accepted mathematical models for such activity.

Mathematical languages for signal processing

Calculus often plays a role in physics based models and general representations for signals.

Probability and statistics are employed for modeling uncertainty associated with unknown factors or random effects like noise. Probability distributions and moments, e.g., variance, and multivariate concepts like correlation matrices are frequently employed in modeling signals and analyzing the performance of algorithms.

Computers can only store and process a finite number of signal values, so linear algebra has an important role in development and analysis of signal-processing algorithms. Matrices, vectors, inner products, eigenvectors, bases, and vector spaces are key players in signal processing.

The real world

Mathematical models are intended to describe signals, but real-world signals are not obligated to follow them. If the assumptions reflected by our models are incorrect, then signal-processing algorithms based on the models may produce poor results.

On the other hand, models do not need to perfectly describe signals to be successful. This is analogous to the situation in modern physics. Newtonian physics does not account for relativistic effects, yet is incredibly useful for describing the behavior of falling objects and the motion of vehicles such as airplanes.

In many cases a simpler model is preferred over a more detailed model both to reduce signal-processing complexity

and increase robustness to deviations between the model and reality. Algorithms that exploit subtle features of detailed models can provide outstanding performance when the model is correct, but fail spectacularly if the model assumptions are violated. It is very important to keep this balance between realism of models and complexity in mind when analyzing data using signal-processing algorithms.

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